

Q1 2021 (01 Sep Shift 2)

Let $f(x)$ be a polynomial of degree 3 such that

$f(k) = -\frac{2}{k}$ for $k = 2, 3, 4, 5$. Then the value of $52 - 10f(10)$ is equal to :

Q2 2021 (31 Aug Shift 2)

Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a function such that

$f(m+n) = f(m) + f(n)$ for every $m, n \in \mathbb{N}$. If $f(6) = 18$,

then $f(2) \cdot f(3)$ is equal to :

- (1) 6
- (2) 54
- (3) 18
- (4) 36

Q3 2021 (31 Aug Shift 2)

Let $S = \{1, 2, 3, 4, 5, 6\}$. Then the probability that a randomly chosen onto function g from S to S satisfies

$g(3) = 2g(1)$ is :

- (1) $\frac{1}{10}$
- (2) $\frac{1}{15}$
- (3) $\frac{1}{5}$
- (4) $\frac{1}{30}$

Q3 (1)

Q1 (26)

$$kf(k) + 2 = \lambda(x-2)(x-3)(x-4)(x-5) \dots (1)$$

put $x = 0$

$$\text{we get } \lambda = \frac{1}{60}$$

Now put λ in equation (1)

$$\Rightarrow kf(k) + 2 = \frac{1}{60}(x-2)(x-3)(x-4)(x-5)$$

Put $x = 10$

$$\Rightarrow 10f(10) + 2 = \frac{1}{60}(8)(7)(6)(5)$$

$$\Rightarrow 52 - 10f(10) = 52 - 26 = 26$$

Q2 (2)

$$f(m+n) = f(m) + f(n)$$

Put $m = 1, n = 1$

$$f(2) = 2f(1)$$

Put $m = 2, n = 1$

$$f(3) = f(2) + f(1) = 3f(1)$$

Put $m = 3, n = 3$

$$f(6) = 2f(3) \Rightarrow f(3) = 9$$

$$\Rightarrow f(1) = 3, f(2) = 6$$

$$f(2) \cdot f(3) = 6 \times 9 = 54$$

Q3 (1) $g(3) = 2$ $g(1)$ can be defined in 3 waysnumber of onto functions in this condition $= 3 \times 4!$ Total number of onto functions $= 6!$

$$\text{Required probability} = \frac{3 \times 4!}{6!} = \frac{1}{10}$$